Brenier Meets Adversarial Training
Optimal Transport Geometry for Robust Learning

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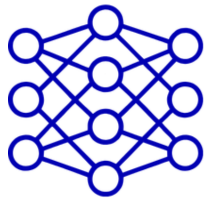
Adversarial Training

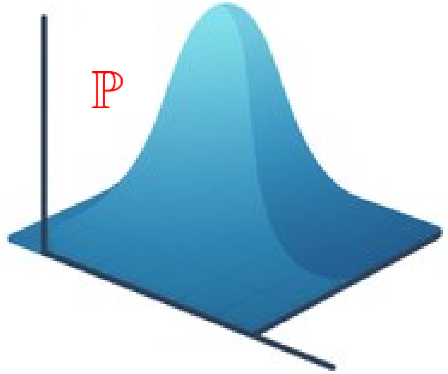
Adversarial Training



Stochastic Programming: $\min_{\theta \in \Theta} \mathbb{E}_{\mathbb{P}}[f(\theta, Z)]$ $\qquad \ell(h_\theta(X), Y)$

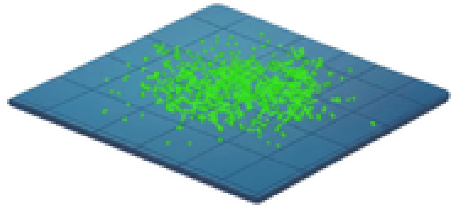
$Z = (X, Y)$ $\qquad X:$

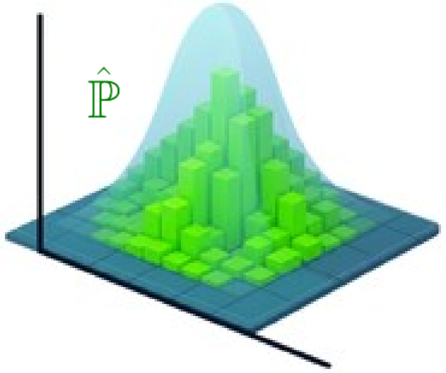
$Y: 1$ (panda)



$\mathbb{P}$

Training Data



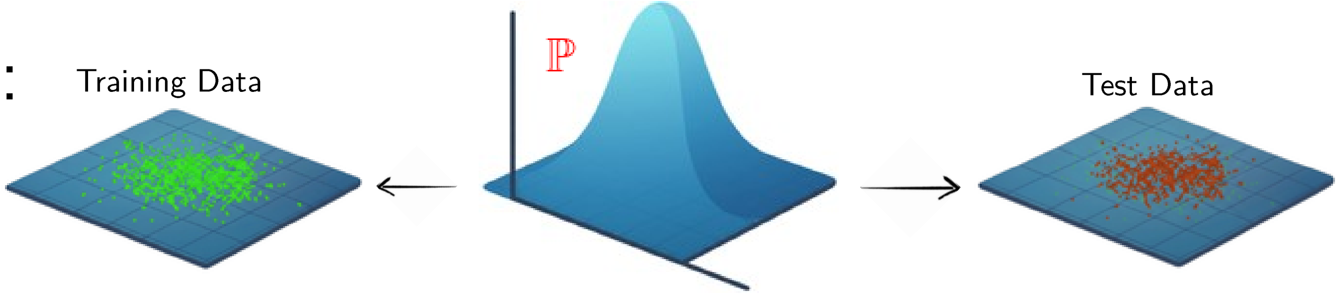


$\hat{\mathbb{P}}$

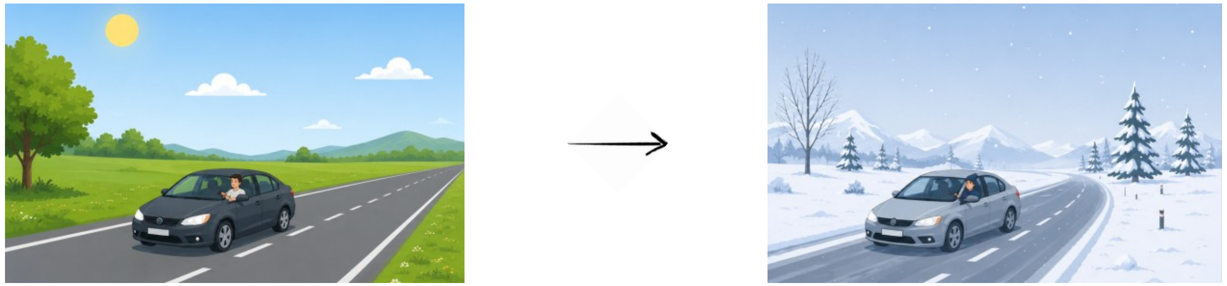
$\longrightarrow$ **ERM:** $\min_{\theta \in \Theta} \mathbb{E}_{\hat{\mathbb{P}}}[f(\theta, Z)] \longrightarrow \theta_{\mathrm{ERM}}$

Adversarial Training: Incentives

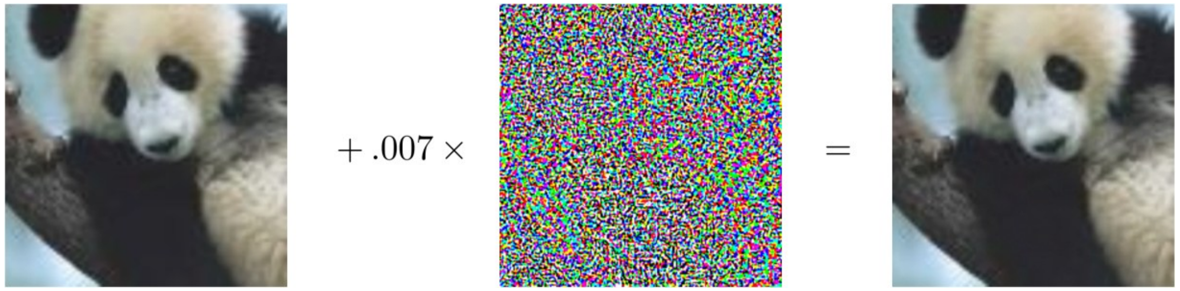
1. Post-decision disappointment:



2. Changes in environment:



3. Adversaries:



x — "panda" — 57.7 % confidence

$\mathrm{sign}(\nabla_x f(\theta, x, y))$

$x + \epsilon\, \mathrm{sign}(\nabla_x f(\theta, x, y))$ — "gibbon" — 99.3 % confidence

Adversarial Training: Formulation

ERM: $\min\limits_{\theta\in\Theta} \mathbb{E}_{\hat{\mathbb{P}}}[f(\theta, Z)]$

Robust Optimization (RO): $\min\limits_{\theta\in\Theta} \mathbb{E}_{\hat{\mathbb{P}}}[\max\{f(\theta, \hat{Z}+\Delta) : \|\Delta\| \leq \varepsilon\}]$

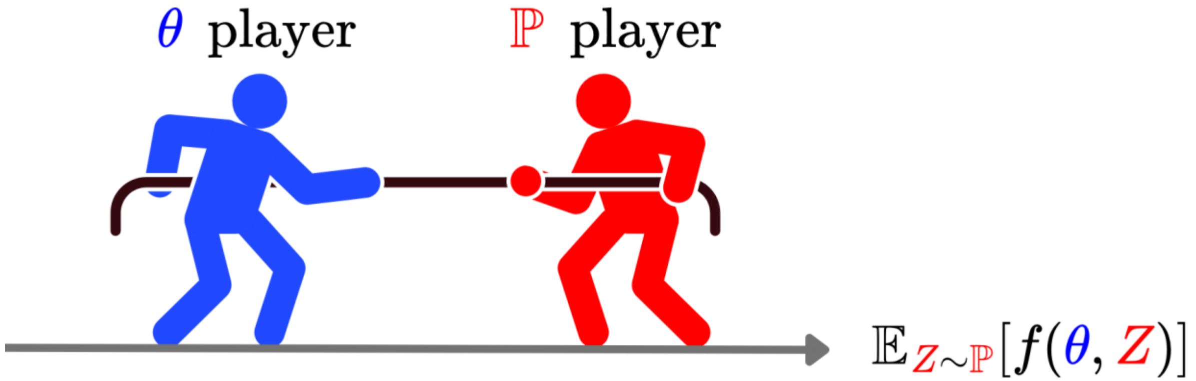
$$\longrightarrow \quad x^{t+1} = \mathrm{Proj}_{\hat{x}+\varepsilon\mathcal{B}_{\|\cdot\|}}\left(x^t + \alpha\,\mathrm{sign}\left(\nabla_x f\left(\theta, x^t, \hat{y}\right)\right)\right)$$

Aleksander Madry et al., ICLR, 2018.

Distributionally Robust Optimization (DRO):

$$\min_{\theta\in\Theta} \sup_{\mathbb{P}\in\boxed{\mathcal{P}}} \mathbb{E}_{Z\sim\mathbb{P}}[f(\theta, Z)]$$

$$\boxed{\mathcal{P} = \mathcal{B}_r(\hat{\mathbb{P}}) = \{\mathbb{P} : \mathrm{W}(\mathbb{P}, \hat{\mathbb{P}}) \leq r\}}$$

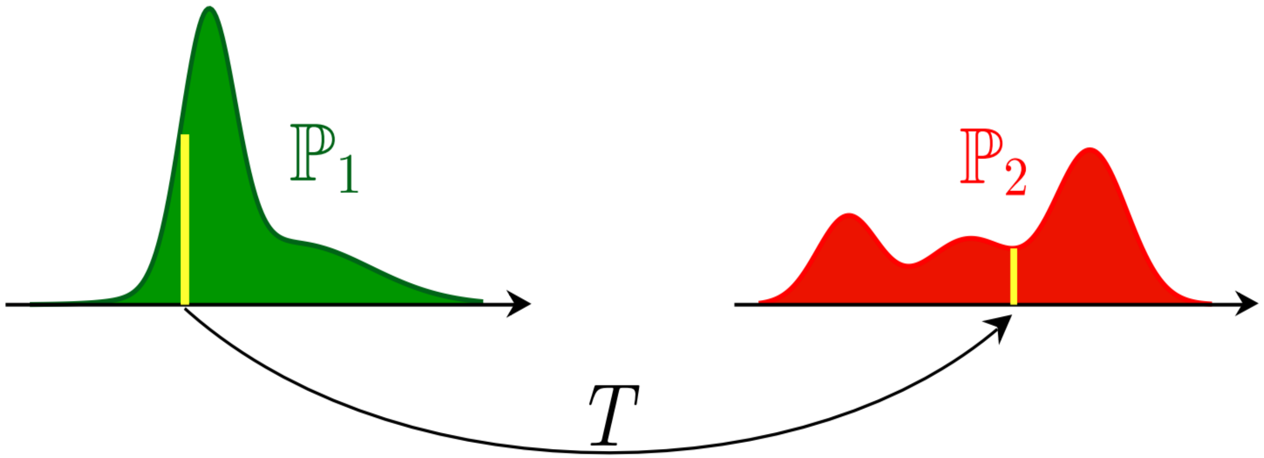


Optimal Transport (OT)

Monge's OT Problem

$$\inf_{T} \int c\left(z_1, T\left(z_1\right)\right) \mathrm{d}\mathbb{P}_1\left(z_1\right)$$
$$\text{s.t.} \quad T_{\#}\mathbb{P}_1 = \mathbb{P}_2$$


Gaspard Monge, 1781



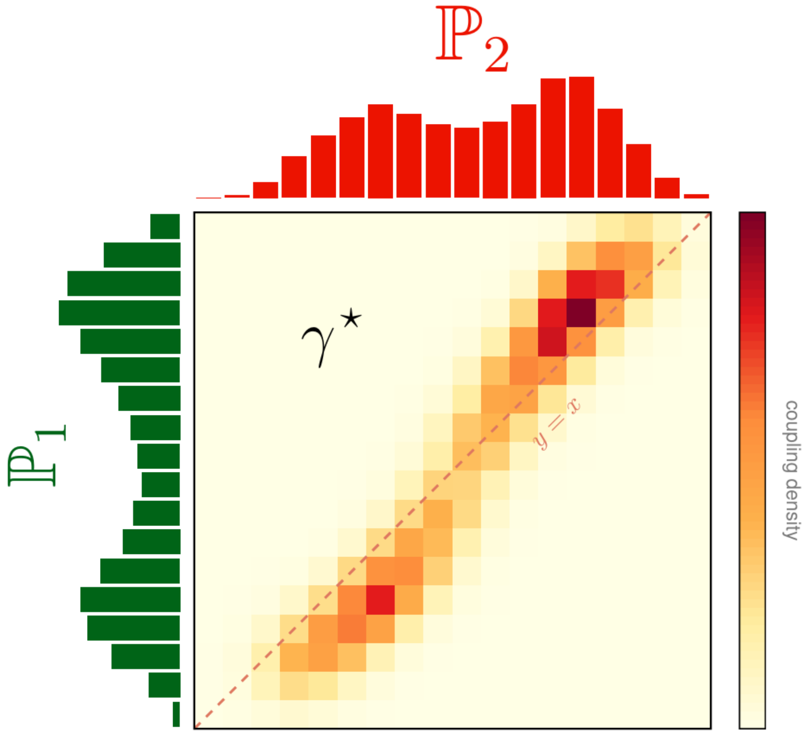
Kantorovich's Relaxation

Relax maps to couplings, mass may split. The problem becomes a linear program, always feasible.

$$\min_{\gamma \in \Pi(\mathbb{P}_1, \mathbb{P}_2)} \int c(z_1, z_2) \mathrm{d}\gamma(z_1, z_2)$$

$$\Pi(\mathbb{P}_1, \mathbb{P}_2) = \left\{ \gamma : (\pi_1)_{\#}\gamma = \mathbb{P}_1, (\pi_2)_{\#}\gamma = \mathbb{P}_2 \right\}$$

Leonid Kantorovich



The Wasserstein Distance

$$\mathsf{W}_p\left(\mathbb{P}_1, \mathbb{P}_2\right) = \left(\min_{\gamma \in \Pi(\mathbb{P}_1, \mathbb{P}_2)} \int d(z_1, z_2)^p \, \mathrm{d}\gamma\left(z_1, z_2\right)\right)^{1/p}$$

This paper: squared-Euclidean cost

$$\mathsf{W}(\mathbb{P}, \widehat{\mathbb{P}})^2 = \min_{\gamma \in \Gamma(\mathbb{P}, \widehat{\mathbb{P}})} \mathbb{E}_\gamma \left[\|Z - \widehat{Z}\|_2^2\right]$$

Leonid Vaseršteĭn

Optimality: Cyclical Monotonicity & Brenier

When is a coupling optimal? Its support must be cyclically monotone.

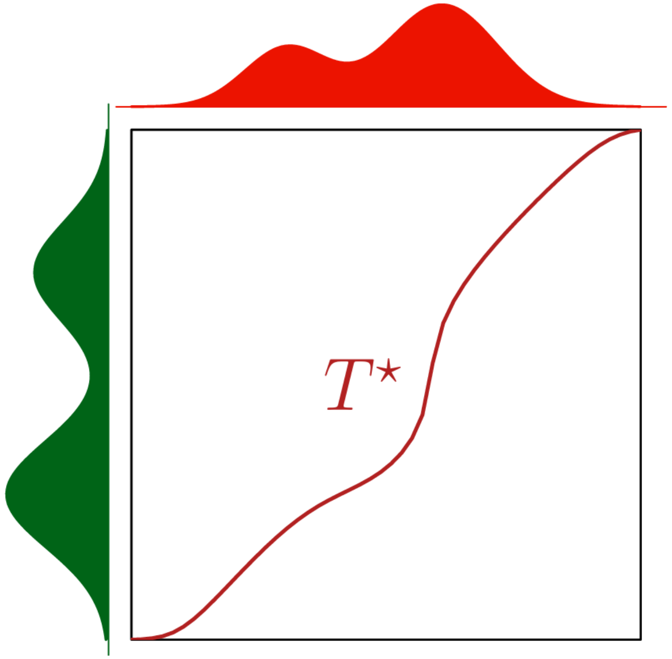
$$\boxed{\sum_{i=1}^{k} \|\hat{z}_i - z_i\|_2^2 \leq \sum_{i=1}^{k} \|\hat{z}_i - z_{\sigma(i)}\|_2^2 \quad \forall \sigma \in \mathcal{S}_k} \iff \sum_{i=1}^{k} \langle z_{\sigma(i)} - z_i, \hat{z}_i \rangle \leq 0$$

no profitable reassignment

$$T \text{ cyc. monotone } \iff T(\hat{z}) \in \partial\psi(\hat{z}), \psi \text{ convex}$$

Brenier's Theorem (1987)

$$c\,(z_1, z_2) = \|z_1 - z_2\|_2^2 \implies T^\star = \nabla\varphi, \quad \varphi \text{ convex}$$

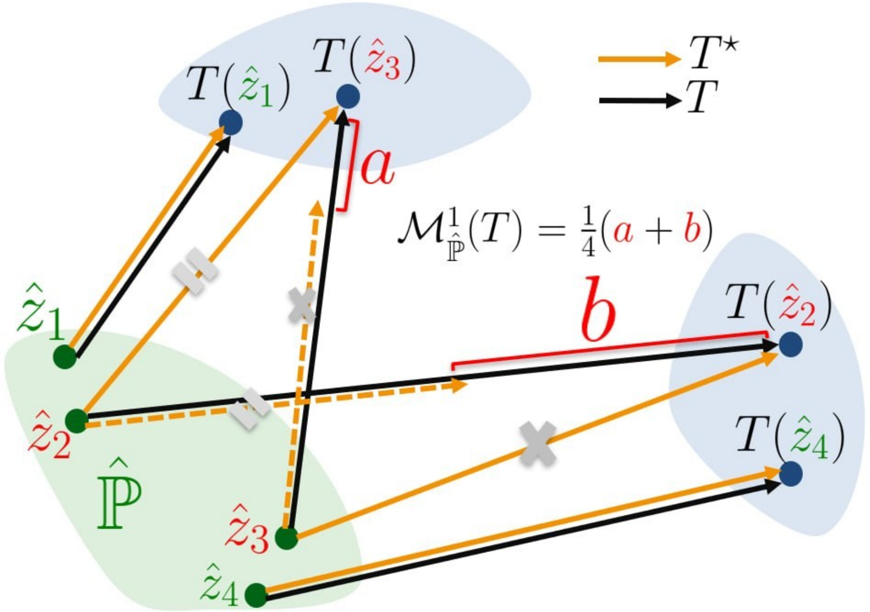


The Monge Gap: Wasteful Transport

Any feasible map that is not cyclically monotone wastes transport budget.
The wasted transport budget is the Monge gap.

$$\mathcal{M}_{\widehat{\mathbb{P}}}(T) = \mathbb{E}_{\widehat{\mathbb{P}}}\left[\|\widehat{Z} - T(\widehat{Z})\|_2^2\right] - \mathsf{W}^2\left(T_\#\widehat{\mathbb{P}}, \widehat{\mathbb{P}}\right)$$

$$\mathcal{M}_{\widehat{\mathbb{P}}}(T) \geq 0 \quad \text{and} \quad \mathcal{M}_{\widehat{\mathbb{P}}}(T) = 0 \iff T \text{ cyc. monotone}$$



Adversarial Risk

DRO: Constrained vs. Regularized

Constrained DRO:

$$\sup \quad \mathbb{E}_{\mathbb{P}}[f(\theta, Z)] = \inf \; \lambda r^2 + \mathbb{E}_{\hat{\mathbb{P}}}\Big[\sup f(\theta, z) - \lambda \|z - \hat{Z}\|_2^2\Big]$$

$\mathbb{P}$: W

Regul

Tractable if the loss function is pointwise maximum of finite concave functions ... Not tractable for Deep NNs.

Peyman Mohajerin Esfahani and Daniel Kuhn, Math. Program., 2018.

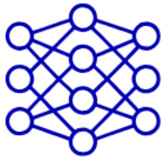
$$= \mathbb{E}_{\hat{\mathbb{P}}}\Big[\max_{z \in \mathbb{R}^m} \{f(\theta, z) - \lambda \|z - \hat{Z}\|_2^2\}\Big] \longrightarrow \text{Point-wis Form}$$

$$= \max_{T \in \mathcal{T}} \; \mathbb{E}_{\hat{\mathbb{P}}}[f(\theta, T(\hat{Z})) - \lambda \|T(\hat{Z}) - \hat{Z}\|_2^2] \longrightarrow \text{Map-based Form}$$

General Recipe for Adversarial Training

$$\min_{\theta\in\Theta} F(\theta) = \min_{\theta\in\Theta}\max_{T\in\mathcal{T}} \mathbb{E}_{\hat{\mathbb{P}}}\left[f(\theta, T(\hat{Z})) - \lambda\|T(\hat{Z}) - \hat{Z}\|_2^2\right]$$

Learner (minimize $F(\theta)$)



parameter θ_t

current parameter θ_t

adversarial map T_t

Adversary (inexact oracle)

returns map $T_t \in \mathcal{T}$

$$\varepsilon(\theta_t, T_t) = F(\theta_t) - \mathbb{E}_{\hat{\mathbb{P}}}\left[f(\theta_t, T_t(\hat{Z})) - \lambda\|T_t(\hat{Z}) - \hat{Z}\|_2^2\right] \geq 0$$

Learner update (projected GD)

$$\theta_{t+1} = \mathrm{Proj}_{\Theta}(\theta_t - \alpha g_t), \qquad g_t = \mathbb{E}_{\hat{\mathbb{P}}}\left[\nabla_\theta f(\theta_t, T_t(\hat{Z}))\right]$$

adversarial error

$\varepsilon(\theta_t, T_t) \geq 0$

Convergence Guarantee:

$$F\left(\bar{\theta}_H\right) - F\left(\theta^\star\right) \leq \frac{L_\theta \|\theta_0 - \theta^\star\|_2}{\sqrt{H}} + \frac{1}{H}\sum_{t=0}^{H-1} \underset{\lambda \mathcal{M}_{\hat{\mathbb{P}}}(T)}{\underset{\vee\!|}{\varepsilon\left(\theta_t, T_t\right)}}$$

Implicit Adversarial Maps

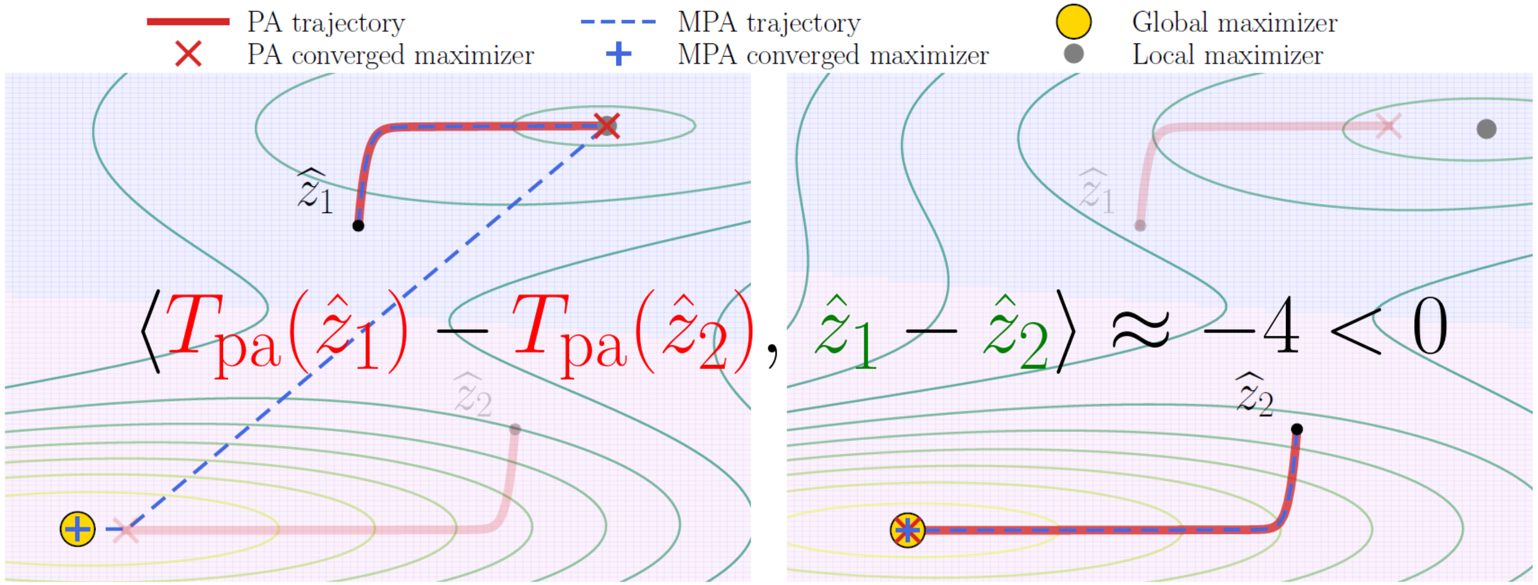
Particle Ascent (PA)

$$F(\theta) := \max_{\mathbb{P}\in\mathcal{P}} \mathbb{E}_{\mathbb{P}}[f(\theta, Z)] - \lambda \mathrm{W}^2(\mathbb{P}, \hat{\mathbb{P}}) = \mathbb{E}_{\hat{\mathbb{P}}}\left[\boxed{\max_{z\in\mathbb{R}^m} \{ f(\theta, z) - \lambda \|z - \hat{Z}\|_2^2 \}}\right]$$

$$\boxed{\begin{aligned} &T_{\mathrm{pa}}(0, \hat{z}) = \hat{z} \\ &T_{\mathrm{pa}}(t+1, \hat{z}) = T_{\mathrm{pa}}(t, \hat{z}) + \eta\Big(\nabla_z f(\theta, T_{\mathrm{pa}}(t, \hat{z})) - 2\lambda\big(T_{\mathrm{pa}}(t, \hat{z}) - \hat{z}\big)\Big) \end{aligned}}$$

Lemma:

if $m = 1$, $f(\theta,$ ... ically monotone.



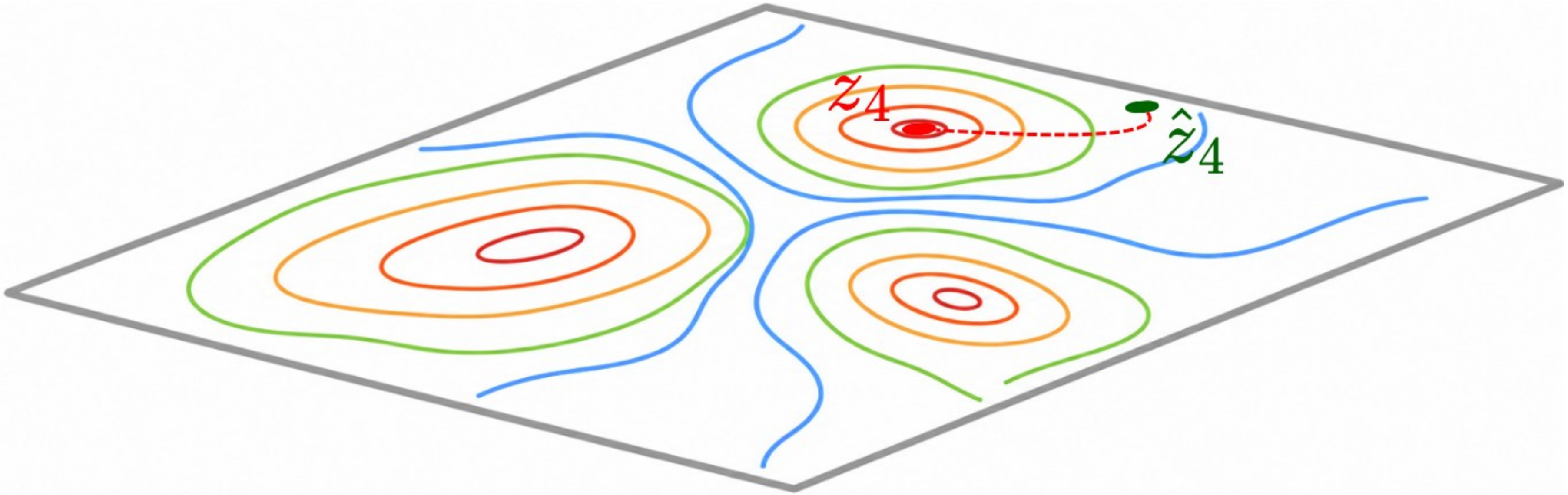
$$\langle T_{\mathrm{pa}}(\hat{z}_1) - T_{\mathrm{pa}}(\hat{z}_2), \hat{z}_1 - \hat{z}_2 \rangle \approx -4 < 0$$

Multi-Start Particle Ascent (MPA)

$$\mathbb{E}_{\hat{\mathbb{P}}}\left[\max_{z\in\mathbb{R}^m}\left\{f(\theta, z)-\lambda\|z-\hat{Z}\|_2^2\right\}\right]$$

$$\{\hat{z}_i\}_{i=1}^B \overset{iid}{\sim} \hat{\mathbb{P}}$$

$$g_4(z) = f(\theta, z) - \lambda\|z - \hat{z}_4\|_2^2 \xrightarrow{\text{PA}} z_4$$

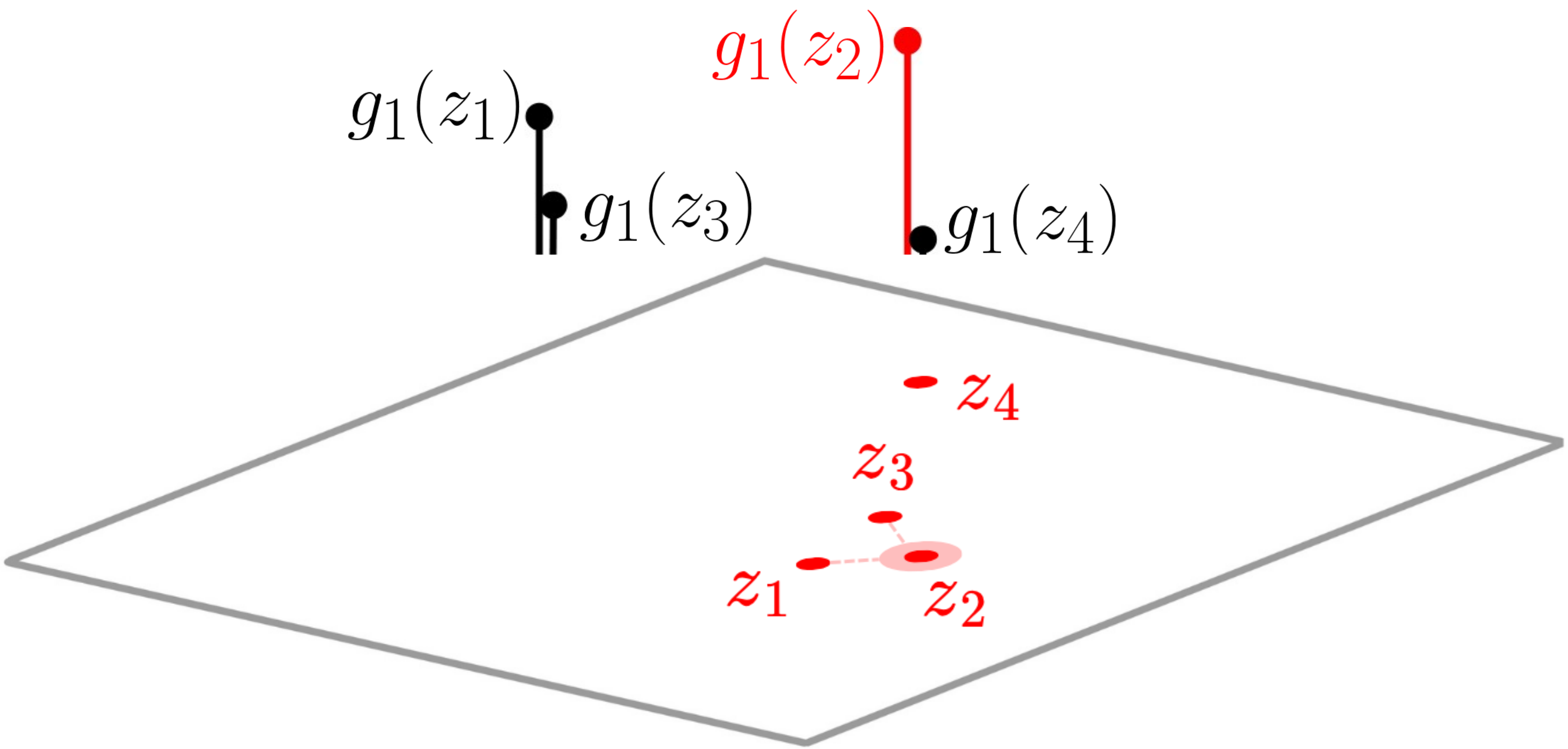


Multi-Start Particle Ascent (MPA)

$$\mathbb{E}_{\hat{\mathbb{P}}}\left[\max_{z\in\mathbb{R}^m}\{f(\theta,z)-\lambda\|z-\hat{Z}\|_2^2\}\right]$$

$$\{\hat{z}_i\}_{i=1}^{B}$$

$$g_1(z) = f(\theta, z) - \lambda\|z - \hat{z}_1\|_2^2$$



Multi-Start Particle Ascent (MPA)

Algorithm 1: Training with implicit adversarial maps

Require: distribution $\hat{\mathbb{P}}$; batch size B; initial iterate θ; step sizes α, η; # epochs T; # inner steps K; # rounds R ($R{=}1$: PA; $R{>}1$: MPA)

  for $t \in [T]$ **do**

    Sample $\{\hat{z}_i\}_{i=1}^{B} \overset{iid}{\sim} \hat{\mathbb{P}}$, and set $z_i \leftarrow \hat{z}_i$ for all $i \in [B]$

    for $r \in [R]$ **do**

      do K **steps of** $z_i \leftarrow z_i + \eta\left(\nabla_z f(\theta, z_i) - 2\lambda(z_i - \hat{z}_i)\right)$ for all $i \in [B]$

      $\mathcal{Z} \leftarrow \{z_i\}_{i=1}^{B}$

      $z_i \leftarrow \arg\max_{z \in \mathcal{Z}} f(\theta, z) - \lambda\|z - \hat{z}_i\|_2^2$ for all $i \in [B]$

    end for

    $\theta \leftarrow \mathrm{Proj}_{\Theta}\left(\theta - \frac{\alpha}{B}\sum_{i=1}^{B} \nabla_\theta f(\theta, z_i)\right)$

  end for

Ensure: θ

Lemma: T_{mpa} is cyclically monotone on $\{\widehat{z}_i\}_{i=1}^{B}$, therefore induces no excess transport cost across the batch.

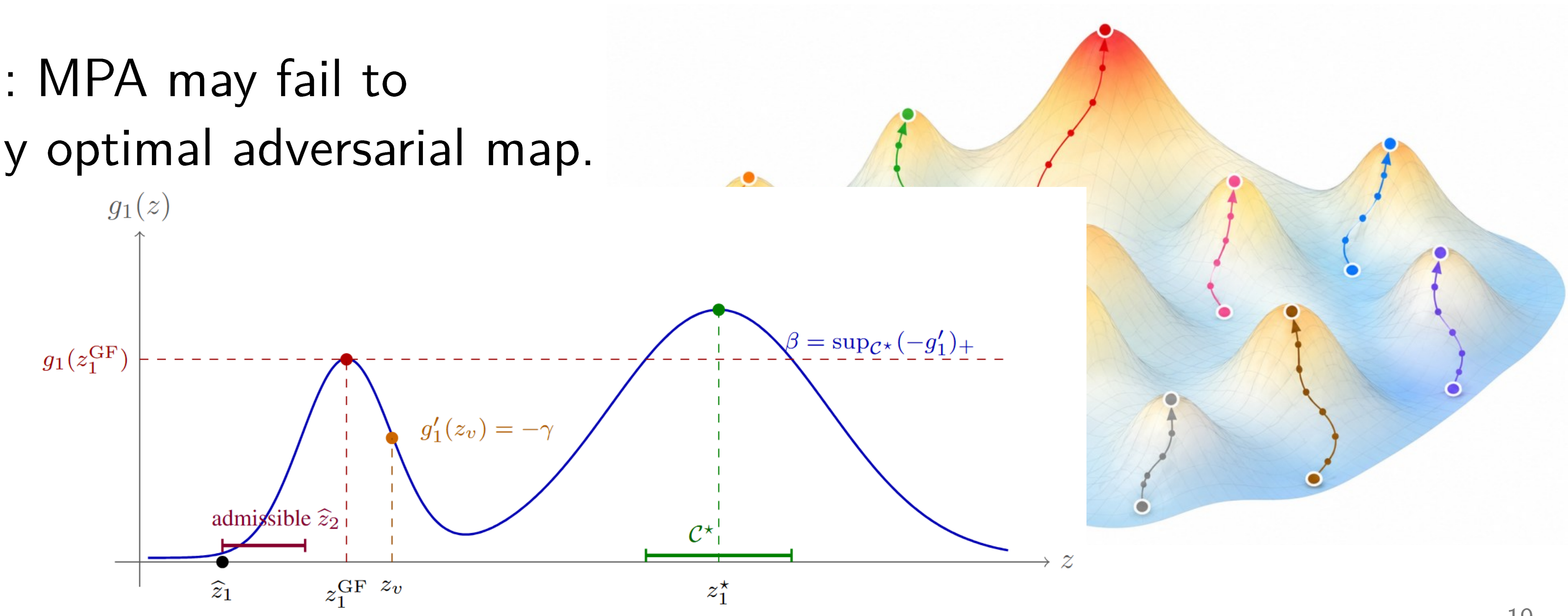
Multi-Start Particle Ascent (MPA)

Three regimes:

1. $\lambda \geq L_{zz}/2$: MPA is equivalent to PA and can find the global maximizer.

2. $\lambda \to 0$: MPA behaves like a multi-start heuristic in global optimization.

3. λ moderate : MPA may fail to find the globally optimal adversarial map.



Explicit Adversarial Maps

Explicit Adversarial Maps

$$F(\theta) = \max_{T \in \mathcal{T}} \mathbb{E}_{\widehat{\mathbb{P}}} \left[f(\theta, T(\widehat{Z})) - \lambda \| T(\widehat{Z}) - \widehat{Z} \|_2^2 \right]$$

$$T \doteq T_\omega$$ Restrict the Adversary to an ***explicit*** parametric family

1

$$\mathcal{L}(\theta; \omega) := \mathbb{E}_{\widehat{\mathbb{P}}} \left[f\left(\theta, T_\omega(\widehat{Z})\right) - \lambda \left\| T_\omega(\widehat{Z}) - \widehat{Z} \right\|_2^2 \right]$$

2

alternate: ascend the map (ω), descend the model (θ)

Algorithm 2: Training with explicit adversarial maps

Input : distribution $\widehat{\mathbb{P}}$; batch size B; initial iterates θ, ω; step sizes α, η; epochs T; inner steps K

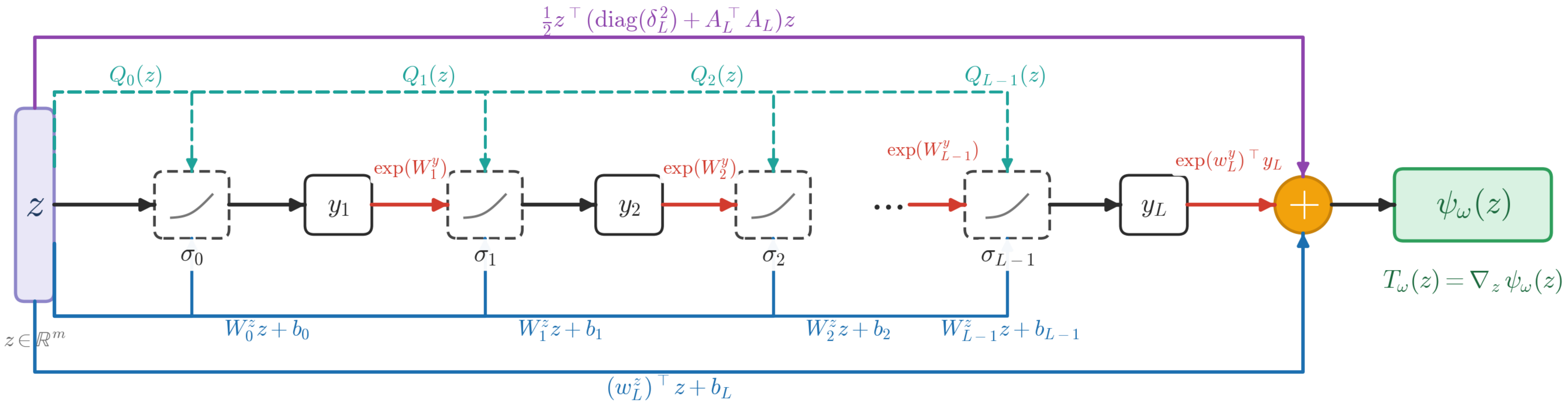
for $t \in [T]$ **do**

- Sample $\{\widehat{z}_i\}_{i=1}^B \overset{\text{iid}}{\sim} \widehat{\mathbb{P}}$
- do K steps of $\omega \leftarrow \omega + \eta \nabla_\omega \mathcal{L}(\theta; \omega)$ inner: ascend the map
- $z_i \leftarrow T_\omega(\widehat{z}_i)$ for all $i \in [B]$
- $\theta \leftarrow \mathrm{Proj}_\Theta\left(\theta - \frac{\alpha}{B} \sum_{i=1}^B \nabla_\theta f(\theta, z_i)\right)$ (outer: Danskin)

Output: θ

Input Convex Neural Networks (ICNNs)

Model a scalar convex potential $\quad T_\omega = \nabla_z \psi_\omega, \quad \psi_\omega : \mathbb{R}^m \to \mathbb{R}$



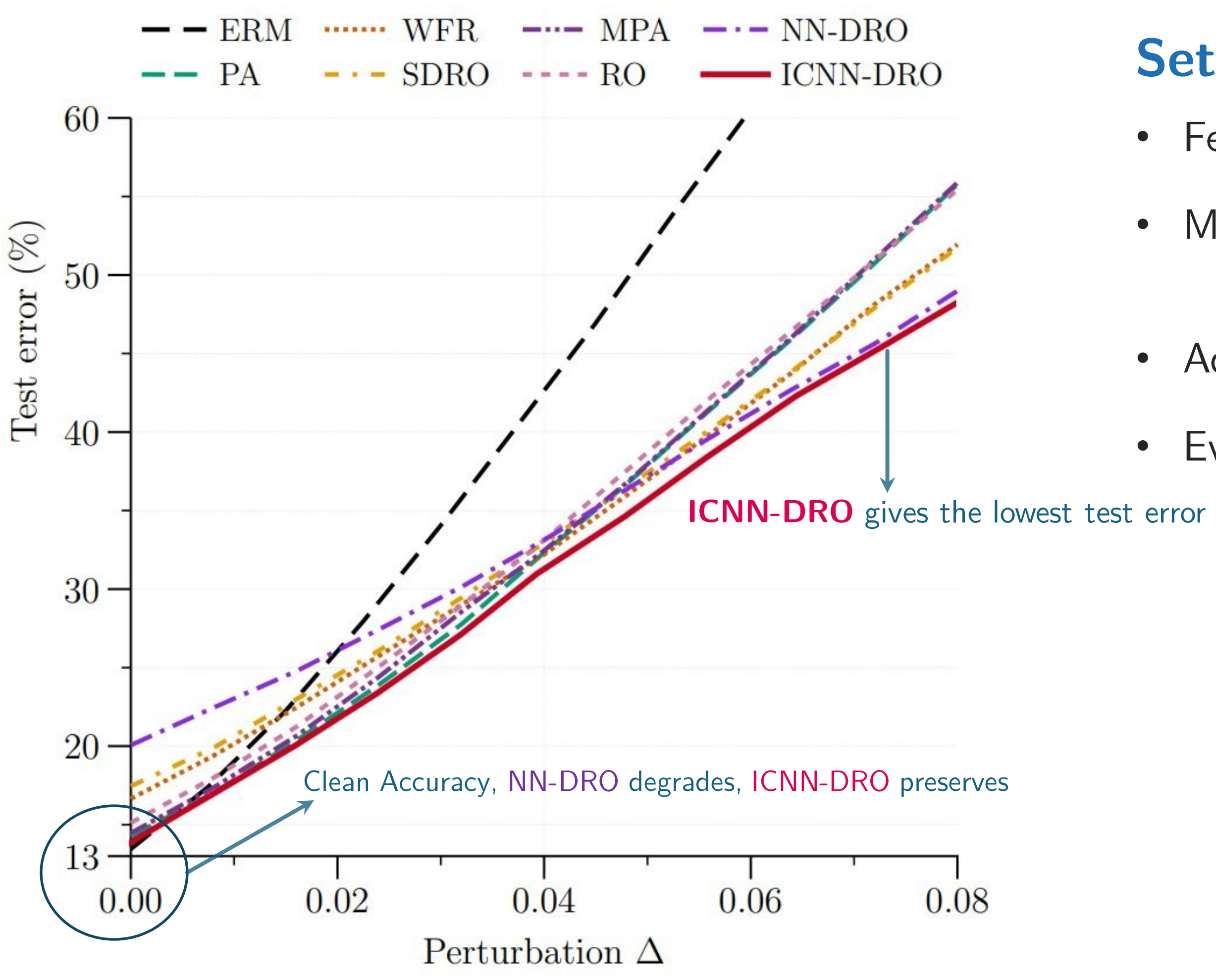
$$y_1 = \sigma(Q_0(z) + W_0^z z + b_0)$$

$$y_{\ell+1} = \sigma\big(\exp(W_\ell^y)\, y_\ell + Q_\ell(z) + W_\ell^z z + b_\ell\big)$$

$$\psi_\omega(z) = \exp(w_L^y)^\top y_L + \tfrac{1}{2} z^\top (\mathrm{diag}(\delta_L^2) + A_L^\top A_L) z + (w_L^z)^\top z + b_L$$

Experiments

Adversarial Logistic Regression on CIFAR-10



Test error vs. relative ℓ_2-PGD attack budget Δ

Setup

- Features $\boldsymbol{x}$ from a frozen ResNet-50 (ImageNet), $\boldsymbol{x} \in \mathbb{R}^{m-1}$
- Multi-class logistic regression with the NLL loss

$$f(\theta, (x, y)) = -y^\top \theta^\top x + \log\left(\mathbf{1}^\top \exp(\theta^\top x)\right)$$

- Adversary perturbs only the features $\boldsymbol{x}$; labels $\boldsymbol{y}$ stay fixed
- Evaluate with an ℓ_2-PGD attack, sweeping budget $\Delta \in [0, 0.08]$

Image-Space Robustness on CIFAR-10

	Adversarial attacks			Unseen domains		Image corruptions					
Method	Clean	$PGD_{\varepsilon=0.5}$	$AA_{\varepsilon=0.5}$	CIFAR-10.1	CIFAR-10.2	1	2	3	4	5	Avg
ERM	**93.12**	14.76	00.20	**87.12**	**83.14**	**91.50**	**87.95**	83.84	77.89	68.41	81.92
PA	90.08	52.41	48.86	80.27	76.16	87.11	84.35	80.21	74.62	69.74	79.20
RO	90.05	53.37	50.26	80.28	76.81	87.15	85.06	81.75	75.35	70.12	79.89
WFR	91.03	53.28	50.37	81.24	77.09	88.60	85.34	82.93	77.56	71.09	81.10
SDRO	90.00	53.12	49.61	80.20	76.48	87.12	84.82	80.33	74.70	70.09	79.41
NN-DRO	89.46	50.34	44.27	80.01	75.71	85.90	82.23	78.55	72.20	67.42	77.26
MPA	91.35	55.00	53.50	82.00	78.32	88.43	85.12	82.61	77.34	70.95	81.05
ICNN-DRO	91.37	**57.36**	**56.65**	82.50	79.30	89.24	86.82	**84.06**	**79.95**	**73.99**	**82.82**

Adversarial attacks

PGD (57.4%) and **AutoAttack (56.7%)**. The **tight** 0.71% PGD–AA gap signals genuine robustness, **not gradient masking**; clean accuracy stays within 1.75% of ERM.

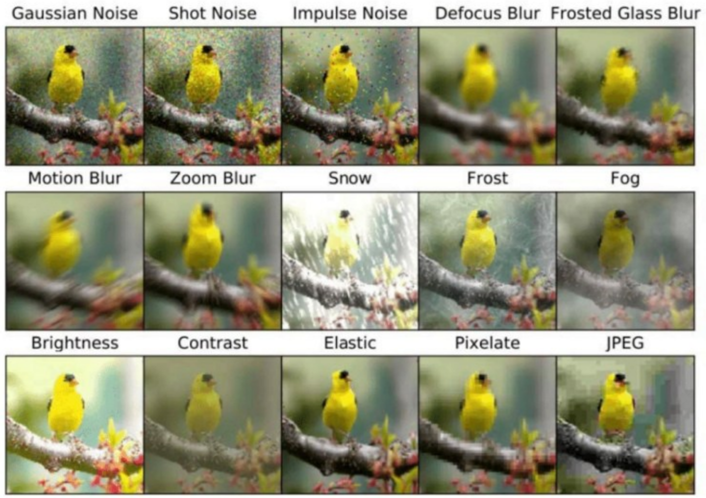
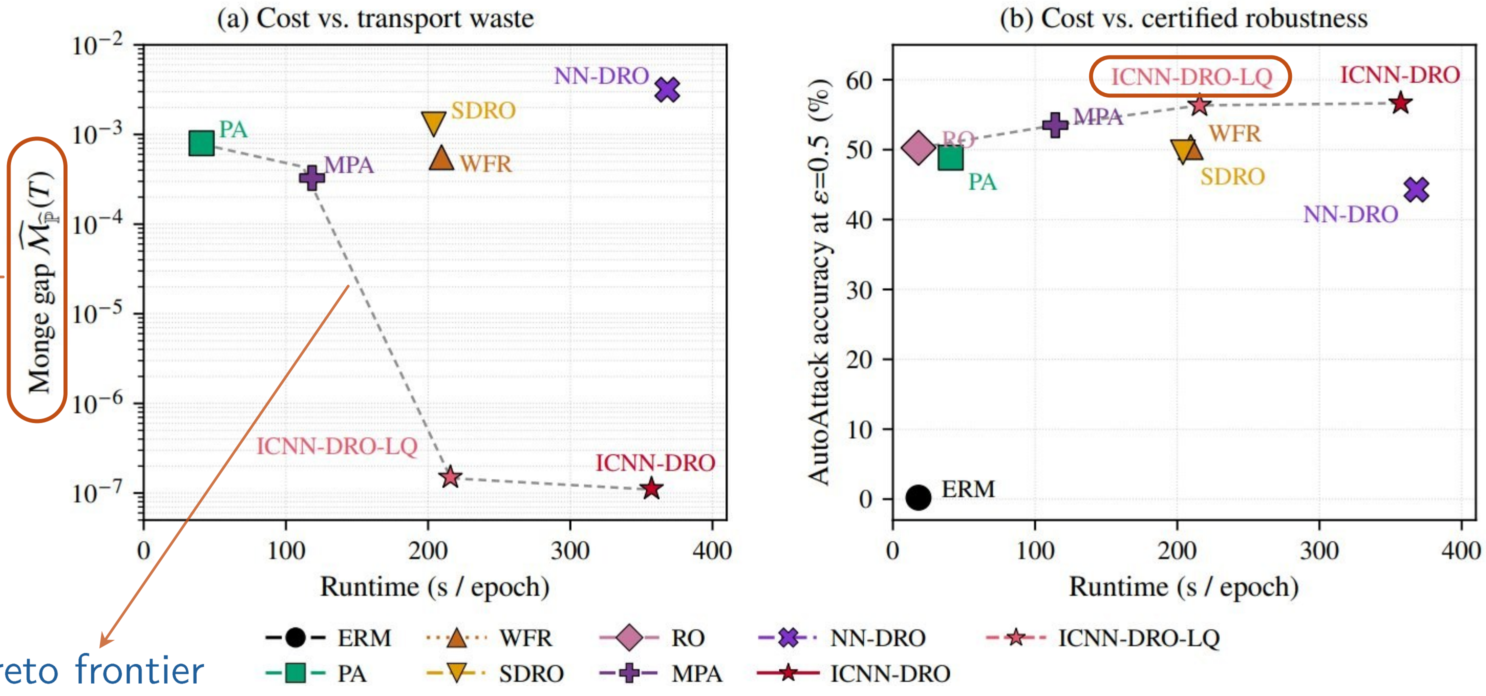


Image corruptions

Best **CIFAR-10-C average (82.8%)**, exceeding even ERM. The margin widens at the harshest severity 5: **74.0% vs 69.7%** for PA.

Runtime vs. Robust Accuracy / Monge Gap

Monge gap ↔ robustness



One Pareto frontier

PA → MPA → ICNN-DRO-LQ → ICNN-DRO

ICNN-DRO-LQ: cheaper

Keeps the quadratic injection **only at the readout layer** (none in the hidden layers). It matches ICNN-DRO's robustness at **≈ half the per-epoch runtime** (216s vs 357s).

Robust Control: Cart-Pole

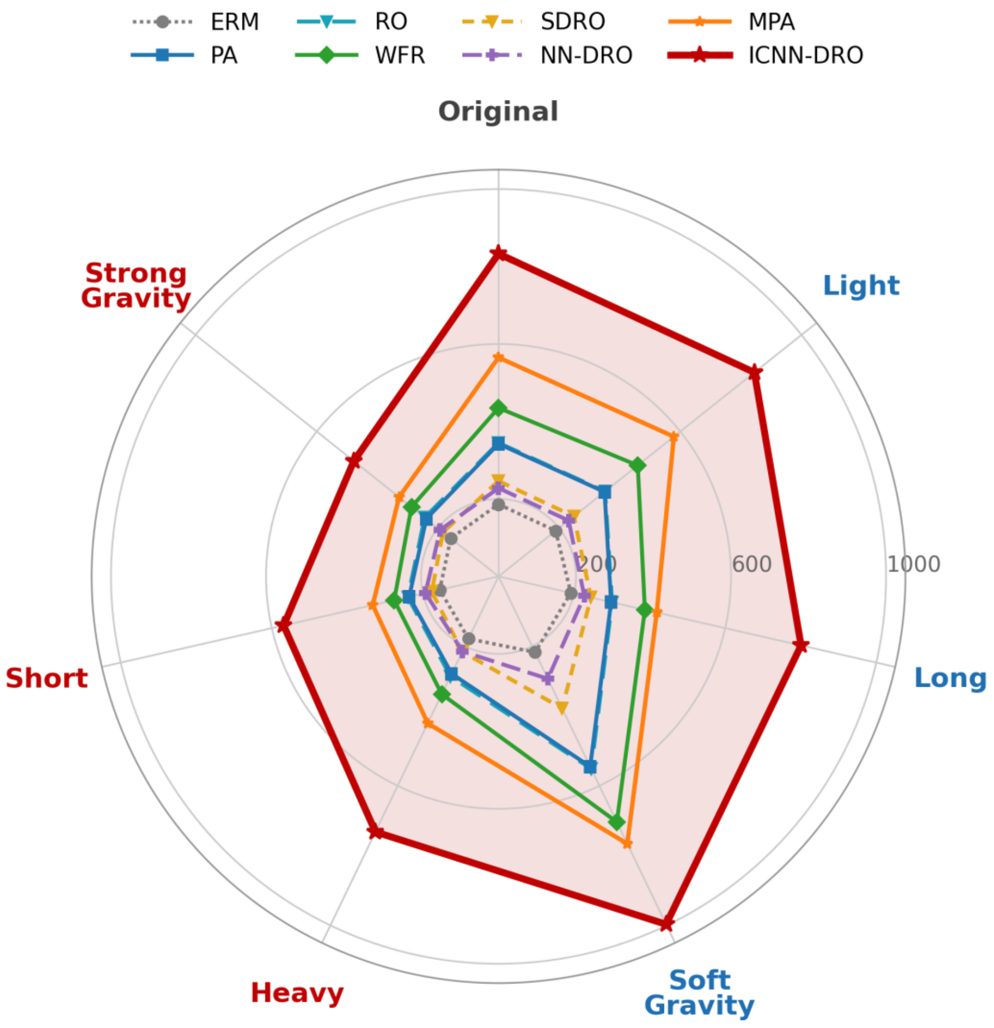
Setup

- Balance a pole on a cart; uncertain **pole mass & length** $z = (m_p, \ell)$ drawn uniformly from $\mathcal{Z} = [0.05, 0.2] \times [0.25, 0.75]$
  Policy π_θ trained with PPO
- Loss = negative expected return (episode length, max 1000):

$$f(\theta, z) := -\mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{H} r(s_t, a_t)\right], \quad z = (m_p, \ell)$$

What the radar shows ($\lambda = 1$)

- **ICNN-DRO** (red, filled) achieves the **longest episode on every variant**
- Largest gains on the **extrapolative harder shifts** (Heavy, Short, Strong Gravity) that lie outside the training box
- **MPA** is the consistent runner-up and clearly beats PA; **WFR** leads the rest but degrades on the hard variants



Thanks for your attention!